

What Is Asthma?

Source: National Heart, Lung, and Blood Institute (NHLBI), National Institutes of Health. NIH Publication No. 20-HL-8121, March 2020. U.S. Government work, public domain.

We know a lot more about asthma today than we did just a decade ago, and we have a much better understanding of how to treat it.

Basic Facts About Asthma

- Asthma is a lung disease. It's a physical and medical problem that needs treatment — it's not something imagined or made up. It's in the lungs, and it's real.
- Asthma is serious. A person can die during an asthma attack, which is why knowing how to take care of your asthma and when to get emergency help is so important.
- Asthma doesn't go away and it can't be cured. Once you develop asthma, you're likely to have it for a lifetime. Even when you feel fine, the asthma is still there and can flare up at any time.
- Asthma can be managed. Like diabetes and high blood pressure, asthma takes ongoing monitoring and management to keep it under control.
- Asthma symptoms result from ongoing inflammation (swelling) that makes the airways very sensitive and narrower than normal. Inflammation protects our bodies, but it can also be harmful when it occurs at the wrong time or stays around after it's not needed.
- The symptoms of asthma are different for different people, and can change over time.

People who have asthma often say it feels like breathing through a straw.

Symptoms

Most people who have asthma have one or more of these symptoms:

- Coughing. Coughing from asthma is often worse at night, making it hard to sleep. Sometimes coughing brings up mucus.
- Wheezing. Wheezing is a whistling or squeaky sound when you breathe.
- Chest tightness. This can feel like something is squeezing or sitting on your chest.
- Shortness of breath. Some people say they can't catch their breath, or feel like they can't get enough air out of their lungs.

How Asthma Affects Your Airways

The airways in your lungs are very sensitive to substances such as tobacco smoke, dust, chemicals, and pollen, or to getting a cold or the flu. These are called asthma “triggers,” because your immune system overreacts to them by triggering the release of cells and chemicals, which cause:

- The inner linings of the airways to become more inflamed (swollen), leaving even less room for air to move through.

- The muscles surrounding the airways to get bigger and tighten, squeezing the airways and making them smaller (called bronchospasm).
- Glands in the airways to produce lots of thick mucus, which further blocks the airways.

These changes can make it harder to breathe. They also can make you cough, wheeze, and feel short of breath. If asthma-related inflammation isn't treated, each exposure to your triggers increases the inflammation, and symptoms are likely to get worse. Normally, your airway is open, so air moves in and out of your lungs freely; when exposed to triggers, the sides of your airways swell and the muscles tighten, leaving less room for air to move freely.

Managing Your Asthma

Discuss your asthma with your health care provider. Together, you can create a treatment plan that will help you:

- Reduce impairment — so you can keep asthma symptoms away, keep up with your usual daily activities, and sleep through the night.
- Reduce risk — so you can prevent asthma attacks, stay out of the emergency room or hospital, and have fewer side effects from your medicines.

Managing your asthma means working with your health care provider, taking your medicines as prescribed, avoiding your triggers, and watching for any changes in your asthma. For more information and resources on lung health, visit NHLBI's Learn More Breathe Better program at nhlbi.nih.gov/BreatheBetter.